

Name: Patrick Flores

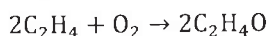
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CHEMICAL REACTION ENGINEERING

FINAL EXAM

CHE 321, CLOSED BOOK, TIME: 120 MIN

1. Ethylene oxide C_2H_4O is produced isothermally at $300^\circ C$ by the vapor-phase catalytic oxidation of ethylene C_2H_4 with air.



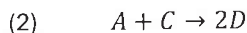
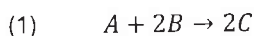
$$r'_{C_2H_4} = -k P_{C_2H_4}^{1/3} P_{O_2}^{2/3}$$

$$k = 4 \times 10^{-8} \frac{\text{mol}}{\text{Pa} \cdot \text{kg} \cdot \text{s}} \text{ at } 300^\circ C$$

The process takes place in two sequential fluidized bed reactors. The feed contains ethylene, oxygen, and nitrogen (inert) with molar flow rates of 2, 2, and $6 \frac{\text{mol}}{\text{s}}$. The pressure of the feed, the first reactor, and the second reactor are maintained at 10, 6, and 4 bar respectively ($1 \text{ bar} = 10^5 \text{ Pa}$).

Estimate the catalyst weight in each reactor, given that the ethylene conversion is 50% after the first reactor and 80% after the second.

2. The elementary (liquid)-phase reactions



are carried out in a 50 dm^3 well-mixed CSTR equipped with a heating jacket. A mixture of A and B is fed to the reactor at the volumetric flow rate of $1 \frac{\text{dm}^3}{\text{s}}$, temperature of 500 K and concentrations $C_{A0} = 1.0 \frac{\text{mol}}{\text{dm}^3}$ and $C_{B0} = 2.0 \frac{\text{mol}}{\text{dm}^3}$. At steady state, the outlet temperature is measured to be 450 K and the outlet concentrations are $C_A = 0.25 \frac{\text{mol}}{\text{dm}^3}$, $C_B = 1.0 \frac{\text{mol}}{\text{dm}^3}$, $C_C = 0.75 \frac{\text{mol}}{\text{dm}^3}$, and $C_D = 0.5 \frac{\text{mol}}{\text{dm}^3}$.

Additional information:

$$\text{Heat capacities: } C_{PA} = 20 \frac{\text{J}}{\text{mol} \cdot \text{K}}, C_{PB} = 20 \frac{\text{J}}{\text{mol} \cdot \text{K}}, C_{PC} = 30 \frac{\text{J}}{\text{mol} \cdot \text{K}}, C_{PD} = 25 \frac{\text{J}}{\text{mol} \cdot \text{K}}$$

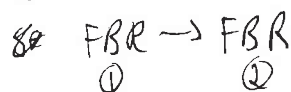
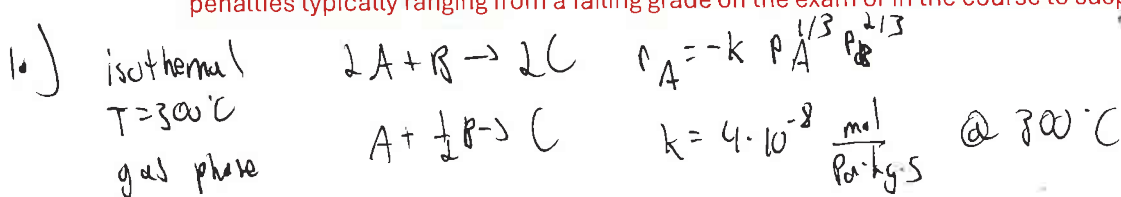
$$\text{Heats of reactions: } \Delta_{r1}H = 20 \frac{\text{kJ}}{\text{mol A}} \text{ and } \Delta_{r2}H = -10 \frac{\text{kJ}}{\text{mol A}}$$

Heating fluid enters the jacket at a temperature $T_{a0} = 600 \text{ K}$, mass flow rate: $\dot{m}_a = 0.02 \frac{\text{kg}}{\text{s}}$ with heat capacity $C_{Pa} = 4500 \frac{\text{J}}{\text{K} \cdot \text{kg}}$

- 2.1. Write the mole balance equations for all four species and estimate the reaction rates $r_{A(1)}$ and $r_{A(2)}$ within the reactor.
- 2.2. Write the energy balance equation for the reactor and estimate the rate of heat transfer into the reactor $\dot{Q}$, assuming negligible shaft work.
- 2.3. Write down the energy balance for the heating fluid and calculate the exit temperature of the heating fluid. Estimate the overall heat transfer coefficient, given the heat transfer area is $A = 120 \text{ dm}^2$.

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feed: A, B, I
 $\uparrow$
 inert is nitrogen

$P_{\text{tot}} = 10 \text{ bar}$ $P_{R1} = 6 \text{ bar}$ $P_{R2} = 4 \text{ bar}$

catalyst weight in $R1, R2 = ?$

$X_1 = 0.50$ $X_2 = 0.80$

$W = \text{catalyst weight}$

knowns

A is C_2H_4

B is O_2

I is N_2

C is C_2H_6O

mole balance

$$F_A|_w - F_A|_{w+\Delta w} + r_A V = 0 \rightarrow \frac{dF_A}{dW} = r_A$$

$$X = \frac{F_{A0} - F_A}{F_{A0}} \rightarrow XF_{A0} = F_{A0} - F_A \rightarrow F_A = F_{A0} - XF_{A0} \rightarrow dF_A = -dXF_{A0}$$

$$\rightarrow \frac{1}{dW} \cdot dF_A = r_A \rightarrow \frac{dX}{dW} = -\frac{r_A}{F_{A0}} \rightarrow \int \frac{-F_{A0}}{r_A} dX = W$$

$$r_A' = (-k) P_{C_2H_4}^{1/3} P_{O_2}^{2/3}$$

for P_{R1}
 inlet mole fractions: $\frac{2}{10} = 0.2 = y_A$
 $0.2 = y_B$
 $\frac{6}{10} = 0.6 = y_I$

$$P_A = P \cdot 0.2 = 6 \text{ bar} \cdot 0.2 = 1.2 \text{ bar}$$

$$P_B = P \cdot 0.2 = 6 \text{ bar} \cdot 0.2 = 1.2 \text{ bar}$$

$$r_A = -\left(4 \cdot 10^{-8} \frac{\text{mol}}{\text{Pa} \cdot \text{kg} \cdot \text{s}}\right) \cdot \frac{10^5 \text{ Pa}}{1 \text{ bar}} \cdot (1.2 \text{ bar})^{1/3} \cdot (1.2 \text{ bar})^{2/3} = -0.0048 \frac{\text{mol}}{\text{kg} \cdot \text{s}}$$

$$W = -\left(\frac{2 \text{ mol/s}}{0.0048 \text{ mol/kg} \cdot \text{s}}\right) (0.50 - 0) = 208.33 \text{ kg}$$

Catalyst weight in 1st reactor = 208.33 kg

2nd reactor $W = \int_{0.5}^{0.8} -\frac{F_A}{r_A} dX \rightarrow F_A = F_{A0}(1-X) = 2(1-0.5) = 1$
 $F_B = F_{B0}(1-X) = 2(1-0.5) \cdot 0.5 = 0.5$
 $F_I = F_{I0} = 6$

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2.2

Energy balance

$$\dot{Q} = F_{A0} \sum_i \theta_i C_{p,i}(T) \left[(\Delta H_1 + \Delta H_2) - \Delta C_p(T) \right] \quad F_{A0} = 0$$

Units: $\frac{J}{s}$, $\frac{mol}{s}$, $\frac{J}{mol \cdot K}$, $\frac{J}{mol}$, $\frac{J}{mol \cdot K}$, $\frac{mol}{s}$, $\frac{J}{mol \cdot K} \cdot K \cdot \frac{mol}{s}$

$$\Delta H_1 = 20 \frac{kJ}{mol} \rightarrow 20000 \frac{J}{mol}$$

$$\Delta H_2 = -10 \frac{kJ}{mol} \rightarrow -10000 \frac{J}{mol}$$

$$\Delta C_p = \Delta C_{p,1} + \Delta C_{p,2}$$

$$\Delta C_{p,1} = 2 C_{p,C} - C_{p,A} - C_{p,B} = 2(30) - 20 - 2(20) = 0$$

$$\Delta C_{p,2} = 2 C_{p,D} - C_{p,A} - C_{p,C} = 2(25) - 20 - 30 = 0$$

$$(\Delta H_1 + \Delta H_2) \cdot F_{A0} = 1(20000 - 10000) = 10000 \frac{J}{s}$$

$$\sum_i \theta_i C_{p,i} \cdot T = \left[\frac{2}{1} \cdot 20 + \frac{1}{1} \cdot 10 \right] \cdot \frac{500K}{(500-250)} = 3000$$

$$30000 \cdot 1 = 30000 \frac{J}{s}$$

$$Q = 30000 \frac{J}{s} + 10000 \frac{J}{s} = 13000 \frac{J}{s} = 13 \frac{kJ}{s} = \dot{Q}$$

2.3

heating fluid energy balance

$$\text{energy in} - \text{energy out} - \text{energy to reactor} = 0$$

$$T_{ao} \cdot m_a \cdot C_{pa}$$

$$600 \cdot 0.02 \cdot 4500$$

$$54000 \rightarrow 54 \text{ kJ}$$

$$T_{at} \cdot m_a \cdot C_{pa}$$

$$T_{at} \cdot 0.02 \cdot 4500$$

$$T_{at} \cdot 0.09$$

$$54 - T_{at} \cdot 0.09 - 13 = 0$$

$$13 \frac{kJ}{s}$$

$$T_{at} = \frac{-13 + 54}{0.01} = 455.55 \text{ K}$$

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